

HW 06 Solutions

1.
$$Q = \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} \\ -\frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ \frac{1}{2} & \frac{\sqrt{2}}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 1-\sqrt{2} & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1+\sqrt{2} \end{bmatrix}$$

MATLAB code follows

(Insert Prob1.txt)

2.
$$Q^T A Q = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

(any order is fine)

MATLAB code follows

(Insert Prob2.txt)

3. (Insert Prob3.txt)